



Progress, Not Regress

A guide to the displays, feelings, measurements, and moral stories through which people decide that something is getting better

A bar is never just a bar

A progress bar seems to report a machine's internal state. Twenty-seven percent is complete; seventy-three percent remains. Yet the bar also manages a human being. It says that the process has not crashed, that waiting has a shape, and that leaving now would waste movement already made. Even when its estimate is imperfect, the display can make an uncertain delay feel bearable.

This section begins with that small interface and expands until it reaches the progress of entire societies. The scale changes, but the questions remain: what is being measured, what cannot be seen, who chose the endpoint, and how does the display influence behaviour? A throbber can hide duration. A placebo button can create a feeling of control without changing a system. Gross domestic product can rise while inequality, health, or environmental conditions worsen. A painting titled *Surely the World Is Growing Better* can use visual irony to challenge the confidence of an age.

Throughout the section, separate three layers. Real progress is a change in the underlying process or condition. Reported progress is the measurement or display of that change. Felt progress is the user's experience of movement, control, and hope. The three can support one another, but they can also diverge. The ethical task is not to eliminate every encouraging design. It is to decide when encouragement helps people understand and complete a worthwhile process, and when it becomes a substitute for truth.

Chapter 1 - Giving waiting a shape

Before graphical interfaces became ordinary, a computer could appear silent while it searched, saved, installed, or calculated. The user had little evidence that work was continuing. A long pause looked much like a frozen machine. The invention of a percentage-done indicator addressed an information problem, but its success depended on psychology as much as computation.

Brad Myers and the value of imperfect information

The official [L007 - history of the progress bar](#) describes research presented by computer science student Brad A. Myers in 1985. In a study of database searches, participants performed tasks with and without a horizontal indicator that filled from left to right. A large majority preferred having the display. The striking result was not that a bar made the computer faster. It made the process more understandable and reassured users that the system had not failed.

A percentage bar needs two estimates: how much work has been completed and how much work exists in total. Both can be uncertain. Copying one hundred files is not necessarily halfway complete after fifty because the remaining files may be much larger. Installing software can reveal new dependencies. A network can slow. Designers often smooth the displayed movement so that a bar does not leap, freeze, or move backward whenever the estimate changes. That makes the experience clearer, but the number is partly a model rather than a direct window into the machine.

The bar carries an implicit promise: the endpoint is known and the process is moving toward it. That promise works well for a download with a known number of bytes. It is less reliable for a security scan, a generative task, or a process that can discover new work. A responsible display therefore matches its confidence to the process. False precision can be more misleading than an honest sign of activity.

Five alternatives, five different claims

An indeterminate progress bar moves without filling toward a known percentage. It says, "the system is active," but not "the system is sixty percent finished." That is honest when total duration cannot be estimated. Its weakness is emotional: endless motion can become evidence that the designer has no useful answer.

A splash screen occupies the interval with branding, artwork, a title, or basic information. It can confirm that an application has launched and create a deliberate first impression. It can also conceal slow startup without telling the user whether work is continuing. If the screen blocks action and offers no status, decoration may increase frustration.

Console output exposes a sequence of operations as text: connecting, loading a module, checking a file, retrying a request. Experts can diagnose exactly where a process is stuck. New users may face a torrent of unexplained language and assume that normal warnings are disasters. Transparency is useful only when the audience can interpret what is shown.

A skeleton screen displays the empty structure of the page that will appear, often as grey blocks where images and text are still loading. It reduces visual surprise and suggests that content will arrive in recognizable places. Unlike a percentage bar, it shows the shape of the destination. Yet it can create a false sense that content is nearly ready when a network request has failed.

A throbber, such as a spinning circle, indicates activity without quantity or endpoint. It is compact and familiar, which makes it useful for short waits. During a long delay, it becomes almost hostile because the user cannot plan. The same symbol that reassures at two seconds frustrates at two minutes.

Which parts of life deserve a bar?

A progress indicator helps when the task has meaningful stages, a partly knowable endpoint, and a person who needs to decide whether to continue or do something else. A visa application benefits from stages such as received, documents checked, decision made, and passport returned. A hospital queue may benefit from information about triage and expected time, but a simple position number could mislead because emergencies rightly change the order.

Some parts of life resist quantification. A friendship at seventy percent, a child who is eighty percent mature, or grief with twelve minutes remaining would pretend that a complex transformation follows a uniform route. The display could pressure people to perform the expected stage rather than experience what is real. Before adding a bar, ask whether the endpoint is shared, whether stages are comparable, and whether the measure will help the person act.

Even a technically accurate bar can exclude. Colour alone may be unreadable to a colour-blind user, motion may trigger discomfort, and a screen reader needs a meaningful text update rather than hundreds of noisy percentage announcements. Direction carries cultural assumptions too: a bar that fills left to right feels natural in some writing systems but not universal. Accessible design exposes the same status through more than one channel and lets users continue other work when the wait is long. Progress information succeeds only when the person who needs it can perceive and use it.

This first chapter establishes the core distinction for the section. A display does not merely describe time. It gives time a story: movement from an incomplete present toward a legible end. The next chapter asks what happens when that story is deliberately more comforting than true.

Sources for this chapter: The official source is [L007 - Gizmodo's account of Brad Myers's 1985 percentage-done research](#). The curriculum's five alternatives are required content even though they have no separate links. Their important difference is epistemic: each claims a different amount of knowledge about activity, duration, stages, or the final state.

Chapter 2 - Designing the feeling of movement

Designers cannot always make a process faster. They can change how much of the delay feels empty, how much control users believe they have, and whether people abandon the task. This can be humane design. It can also become manipulation. Placebo buttons, airport walks, and completion nudges belong together because each changes experience without necessarily changing the underlying duration or goal.

Placebo control

The [L008 - article on placebo buttons](#) describes controls that appear to produce an effect even when they are disabled, automated, or reserved for special access. Common examples include some elevator close-door buttons and pedestrian crossing buttons. The details vary by building, city, time, and mode. It would be wrong to conclude that all such buttons are fake. The curriculum's question is about the psychological and ethical pattern.

People learn that pressing a button produces a result. If the light changes shortly after the press, the user may infer causation even when the signal follows an automatic cycle. The button supplies agency during a wait. That can reduce frustration and discourage unsafe improvisation. Yet it also creates an inaccurate model of the system. A pedestrian who thinks every press changes the light may not understand why accessible signals, timed cycles, or traffic priorities operate differently.

Deception is easier to defend when the apparent control protects a user from unnecessary anxiety, no important decision depends on the belief, and no safer truthful design exists. It is harder to defend when the system shifts blame onto the user, collects consent, extracts money, hides failure, or creates risk. A fake "cancel" button on a dangerous process is not a harmless comfort. Nor is a button that makes people think they have opted out of data collection.

Occupied time and unoccupied time

The official [L009 - Houston airport baggage story](#) reports a counterintuitive intervention. Passengers spent roughly one minute walking to baggage claim and seven minutes waiting at the carousel. Adding baggage handlers reduced the total to an industry-acceptable level, yet complaints continued. The airport then routed passengers so that the walk was much longer. Total time remained similar, but people spent more of it moving and less of it standing at an apparently failed endpoint. Complaints fell sharply.

The story does not prove that walking is always preferable or that airports should waste passengers' energy. A longer route can disadvantage people with disabilities, young children, tight connections, or heavy luggage. Its value is explanatory: occupied time often feels shorter than unoccupied time. Movement supplies evidence that the journey has begun. Waiting at the carousel feels worse because the passenger has arrived at the place where the bag should be and can see no reason for the delay.

Similar designs appear in queues that wind past information, restaurants that let customers watch food preparation, and call centres that offer an estimated wait or a callback. The best intervention can improve both reality and experience. Honest information, a useful activity, a comfortable space, or the ability to leave and return respects the person's time. Making the route needlessly confusing merely to disguise delay uses the same psychology against the user.

Progress as a nudge

Forms, applications, surveys, and learning platforms often show stages or percentages. The [L010 - discussion of progress indicators and engagement](#) connects interface design with the feedback systems of games. A user who sees three of four steps complete may be more willing to finish than one facing an undivided page. The endowed-progress effect can make a person feel that a journey has already begun, while a visible near-end can make abandonment feel costly.

A nudge changes the environment in which people choose without simply forbidding alternatives. Ethical nudging depends on the goal, transparency, reversibility, and interests involved. Helping a student complete a scholarship application they chose is different from making it difficult to cancel a subscription. A progress display becomes a dark

pattern when it pressures people toward an outcome that mainly benefits the designer, hides costs until late in the route, or turns refusal into a maze.

Designers should also ask whether completion is always desirable. A medical questionnaire should encourage accuracy, not speed. A loan application may require moments that invite the borrower to reconsider rather than celebrate ninety-five percent completion. Persuasion can support autonomy when it helps people carry out considered intentions. It undermines autonomy when it exploits attention, confusion, or sunk costs.

The branch is therefore not a contest between neutral interfaces and manipulative ones. Every interface organizes attention. The practical test is whether the design makes the process, stakes, alternatives, and exit clear enough for the user to remain the author of the decision.

Fairness also changes the answer. Occupying a traveller's time may reduce average complaints while imposing a larger burden on a minority. A six-minute walk is exercise for one passenger and a painful barrier for another. A required online form may feel pleasantly short to a fluent reader and exhausting to someone using translation or assistive technology. Designers need segmented evidence, not only one satisfaction score. A humane intervention offers alternatives, such as seating, mobility assistance, saved progress, or a callback, so that the emotional solution does not depend on everyone having the same body and available time.

Sources for this chapter: Use [L008 - "Placebo Buttons"](#) to examine perceived control, but do not generalize every public button from anecdote. The [L009 - airport baggage case](#) illustrates occupied time and should be evaluated for accessibility as well as complaints. The [L010 - UX article](#) opens the nudge question; it is a design discussion, not proof that one indicator produces the same completion effect in every context.

Chapter 3 - Games inside the delay

One response to waiting is to explain it. Another is to occupy it. In 1994, Namco placed a small game inside the loading interval for *Ridge Racer*. Chrome later turned a lost internet connection into a dinosaur's run. Restaurant tablets offer children games while food is prepared, sometimes for a charge that the adult notices only on the bill. These cases ask whether an auxiliary game serves the waiting person or captures them for someone else.

Ridge Racer and a patented distraction

The official [L011 - history of loading-screen minigames](#) explains that the PlayStation version of *Ridge Racer* allowed players to play *Galaxian* while the main game loaded. The choice connected two Namco titles and transformed dead time into play. It also became associated with a United States patent filed in the 1990s and expiring in 2015, a reminder that even a tiny design convention can be shaped by intellectual property.

The minigame changes the experience but not the load. That is not automatically deceptive because the player knows the main game is preparing. The auxiliary activity can preserve immersion and goodwill. Yet it also competes for attention. If the main process becomes ready without a clear signal, the distraction may delay the user rather than help them.

Modern systems use other forms of occupied loading: gameplay tips, lore, practice controls, animated scenes, and interactive tutorials. Each can do useful work during unavoidable delay. The strongest design makes that work optional and relevant. A person who needs a pause should not be forced to perform another task merely because the system cannot load quickly.

The Chrome dinosaur turns failure into a place

Google's [L012 - interview about the Chrome Dino](#) traces the offline game to 2014. The visual joke sends the user back to a time before widespread connectivity. Pressing the space bar turns an error page into an endless runner. Cacti and later obstacles give a disconnected user an immediate, understandable challenge while the network may recover.

The dinosaur is powerful because it does not pretend that the requested page is loading. The error remains visible. The game acknowledges failure and offers a voluntary activity. It gives the waiting interval an identity of its own rather than painting a fake percentage over an unknown outage.

There is still a design tradeoff. An error screen should help a user diagnose the problem, retry, or work offline. Entertainment must not replace recovery tools. In a school or workplace, an endless game can also outlast the interruption that justified it. Good auxiliary design keeps the original goal recoverable.

When waiting becomes a sale

The restaurant case adds money and consent. The [L013 - NBC Chicago report on table-game fees](#) describes tablets at restaurant tables that offered children's games for a fee. Some adults reported noticing the charge only after children had played. A price may have appeared on the screen, but the system did not necessarily require an adult's informed approval.

This is not the same as a free loading minigame. The restaurant creates the waiting context, places an attractive device within a child's reach, and adds a small charge to a larger bill. The customer may not know whether touching a game starts payment. Consent hidden in visual clutter is legally or technically possible without being meaningfully informed.

A fair system would make the price prominent, require adult confirmation, explain whether the fee is per game or per table, and allow the device to be disabled. It should not rely on the low chance that a customer will challenge a small amount. The broader lesson is that "helping people pass time" can become a business model. The designer may have an incentive to preserve the wait, fill it with engagement, and monetize the captive audience.

Waiting can become part of the experience

Not every auxiliary activity is digital. Museums use queue exhibits. Clinics provide clear health information. Theme parks build stories into the line. Restaurants invite customers to watch an open kitchen. These experiences can make waiting informative or social. They can also normalize delays that an organization should fix.

Age and context matter. A child waiting for food may not understand that a brightly animated game initiates a purchase, while an adult facing an airport delay may welcome an optional puzzle. Schools and hospitals carry additional duties because the audience may be captive and attention may be needed for instructions. An auxiliary game should pause instantly when the main task is ready, work without collecting unnecessary data, and avoid rewards designed to keep the user playing after the delay ends. Otherwise the system has quietly changed destinations: from helping a person wait to maximizing engagement.

There is also an opportunity cost. A loading screen can teach controls or explain what the system is doing instead of offering unrelated entertainment. A clinic can tell a patient which documents will be needed. A transit queue can show service updates. The best use of waiting often prepares the next stage, reducing errors and anxiety at the same time. Fun remains valuable, but relevance is one way to keep the side journey from taking over the route.

The right comparison needs two measures. First, did the intervention reduce harm or increase value during an unavoidable interval? Second, did it reduce the organization's motivation to shorten avoidable delay? A game can make a route kinder. It should not become evidence that the slow route no longer matters.

Sources for this chapter: The curriculum's three cases are [L011 - Polygon on *Galaxian* and loading-screen history](#), [L012 - Google's account of the Chrome Dino's creation](#), and [L013 - NBC Chicago on restaurant game fees](#). The first two explain design intent; the third adds the questions of price disclosure and adult consent.

Chapter 4 - A country cannot fit inside one number

A progress bar needs an endpoint. A society needs many. Economic output, income, distribution, health, education, poverty, productivity, freedom of movement, and reported life satisfaction describe different dimensions. None is a neutral answer to "How well are we doing?" Each selects a unit, population, time period, and theory of what should count.

Production and income

Gross domestic product, or GDP, measures the market value of final goods and services produced within a country's borders during a period. "Domestic" makes location central. A foreign-owned factory's production contributes to the host country's GDP. GDP per person divides the total by population and is more useful for rough comparisons of average scale, but it is not the income received by a typical resident and does not reveal distribution.

Gross national income, or GNI, shifts from location to income received by residents. It adds net primary income from abroad to GDP: compensation earned across borders and investment or property income received from non-residents, minus equivalent payments to non-residents. It does not simply add every remittance. Personal transfers are secondary income and enter gross national disposable income rather than GNI, although compensation earned by short-term or border workers can be primary income. GDP and GNI can therefore diverge when cross-border work, ownership, and investment income are important. Neither measure automatically counts unpaid care, leisure, environmental damage, household security, or the value of public services well.

Labour productivity usually compares output with labour input, such as GDP per hour worked or per worker. Higher productivity can support higher wages and living standards because more value is produced from a unit of work. It does not guarantee that gains reach workers, that work is safe, or that output is environmentally sustainable. A metric of efficiency is not a full metric of human purpose.

Distribution and deprivation

The Gini coefficient describes inequality in a distribution, usually income or consumption. A value near zero represents greater equality; a value near one, or one hundred on a different scale, represents greater inequality. Two countries can have the same Gini while having very different average incomes and social conditions. The measure also compresses where inequality sits. A gap at the top can produce the same headline value as hardship concentrated at the bottom.

The Multidimensional Poverty Index, developed through UNDP and the Oxford Poverty and Human Development Initiative, examines overlapping deprivations in health, education, and standard of living. It combines incidence, the share of people classified as multidimensionally poor, with intensity, the average share of weighted deprivations they experience. The [2025 global MPI](#) also connects poverty exposure with climate hazards. Its richness comes with a warning: survey years differ, data can be older than the report year, and weights reflect an explicit methodology.

Capabilities, happiness, and mobility

The Human Development Index combines life expectancy, education, and income into one composite value. It was designed to move debate beyond economic output, but averaging can hide inequality within each dimension. UNDP therefore publishes additional measures and country profiles rather than claiming that one rank tells the whole story.

A happiness index usually relies on survey responses about life evaluation. The World Happiness Report famously uses a ladder question on which respondents rate their lives. Such data give people a voice rather than assuming that income equals wellbeing. Culture, question wording, current events, sampling, and the difference between daily emotion and overall evaluation all matter.

A passport index ranks travel access associated with a passport, often by the number of destinations reachable without obtaining a visa in advance. This measures mobility privilege, diplomatic relations, and border policy. It does not measure whether most citizens can afford travel, whether migrants are treated fairly, or how meaningful a destination is. Competing indexes can use different rules and produce different ranks.

The Big Mac Index, created by *The Economist*, compares the local price of a broadly standardized product as an informal guide to purchasing-power parity. It is memorable because a hamburger bundles local wages, rent, ingredients, taxes, and exchange rates. It is not a precise currency forecast. Product composition, market strategy, regulation, and eating habits vary.

Reading your country's dashboard

When the curriculum asks teams to look up their country, preserve the year and source beside every number. Do not place a 2025 passport rank next to a Gini value derived from a 2018 household survey and call both "current" without explanation. Check whether a ranking is global, whether territories are included, whether values are estimates, and whether higher numbers mean improvement.

Then look for disagreement. GDP per person may rise while reported life evaluation falls. Productivity may grow while the Gini coefficient worsens. HDI can improve while a particular region remains deprived. The disagreement is not statistical failure. It reveals that the country is moving in different directions under different definitions of "there." A good social dashboard keeps those tensions visible.

Composite rankings add another layer of judgment. To combine health, education, and income, analysts must normalize different units and assign a formula or weights. Changing those choices can change the rank even when underlying lives stay the same. Missing data may exclude a country or force estimates. National averages can hide region, gender, ethnicity, disability, age, or migration status. A careful team therefore opens the methodology before celebrating the number and pairs the headline with disaggregated data and testimony from people whose experience the average compresses.

Sources for this chapter: The nine indicators are official curriculum terms even though WSC supplies no direct links for them. Use primary or methodological pages such as the [World Bank GDP data](#), the [World Bank GNI series](#), [UNDP Human Development Index data](#), and [UNDP's 2025 MPI release](#). Rankings from commercial passport indexes and the Big Mac Index must be labelled with their publisher and method. Never compare values without their source year.

Chapter 5 - Why progress is not a law of nature

A filling bar moves in one direction. History does not. The final branch challenges the modern habit of imagining that time itself carries societies toward improvement. It places the Second Law of Thermodynamics beside Martin Luther King Jr.'s moral arc, the historical emergence of belief in progress, and Will Crawford's 1909 painting *Surely the World Is Growing Better*. The comparison is intentionally uncomfortable because physical disorder, moral hope, and social measurement belong to different kinds of claim.

Entropy is not a verdict on society

The official prompt says that closed systems grow more disordered, but the strict distinction matters. In one common formulation, the Second Law says that the entropy of an isolated system does not decrease. A closed system can exchange energy but not matter; an isolated system exchanges neither. Entropy is a technical quantity related to the number and distribution of possible microscopic states, not simply "messiness" or evil. A bedroom can become cleaner because a person supplies energy and exports waste and heat. Earth is not isolated; it receives energy from the Sun and radiates energy into space.

Using entropy as a metaphor can still be useful. Institutions, infrastructure, knowledge, and trust require maintenance. A bridge does not repair itself. A right written in law may be weakened if people stop enforcing it. Yet the physical law does not prove that societies must become worse, any more than local biological growth disproves thermodynamics. Scientific vocabulary should not be stretched into moral inevitability.

The moral arc is an obligation, not a timer

Martin Luther King Jr. used the image of a long moral arc bending toward justice, drawing on nineteenth-century abolitionist minister Theodore Parker. In speeches and sermons, the phrase expressed hope inside organized struggle. Removed from that context, it can sound as if justice arrives automatically when enough time passes. King's life demonstrates the opposite. Laws, institutions, protest, sacrifice, persuasion, and power shape the curve.

Progress can reverse. Democratic rights expand and contract. Public health improves while new risks appear. Technology increases capability without deciding how that capability is used. A society can reduce one injustice and preserve another. This is why a historical graph needs mechanisms, not merely an upward trend line.

The invention of a future expected to improve

The official essay [L014 - "Progress Isn't Natural"](#), preserved at [L015 - its alternate archive link](#), argues that confidence in deliberate human improvement is historically recent. Many earlier worldviews imagined decline from a golden age, recurring cycles, providential history, or knowledge bounded by authoritative ancient texts. Between the era of Columbus and Newton, European thinkers increasingly believed that inquiry and useful knowledge could improve material life.

This claim should not be turned into "earlier people never innovated" or "Europe invented progress alone." Civilizations across the world accumulated knowledge, engineered systems, and debated good government. The narrower argument concerns a widely institutionalized expectation of open-ended, deliberate improvement and the social permission to claim that new knowledge could surpass ancient authority. That expectation supported research, invention, and investment. It also accompanied empire, extraction, and destructive ideas presented as modernization.

The last point changes how progress stories should be judged. A railway, dam, school system, or public-health campaign can deliver real gains while being imposed through coercion or distributing costs along racial and colonial lines. People labelled "backward" were often denied authority over what development meant. A rising national average can coexist with land loss, cultural destruction, or environmental sacrifice concentrated elsewhere. Progress is not disproved by these harms, but its unit of analysis must widen. Ask progress for whom, chosen by whom, compared with which alternative, and reversible at whose request.

Will Crawford's crowded answer

The curriculum links [L016 - Will Crawford's 1909 *Surely the World Is Growing Better*](#). The title invites optimism, but the image operates through irony. Crawford, an American illustrator and cartoonist, fills the scene with signs of conflict, danger, scandal, or social disorder while the confident title hovers as a challenge. The work asks viewers to compare a slogan with the evidence crowded beneath it.

A contemporary version would not need to copy the objects. It might place medical breakthroughs beside unequal access, renewable energy beside rising demand, instant communication beside misinformation, and falling rates of some violence beside war visible in real time. The point would not be that everything is worse. Satire works by exposing the distance between simple confidence and mixed reality.

Progress requires maintenance and choice

The chapter ends where the section began. A progress bar moves because a programmer defined a task and a machine performs it. Social progress has no single programmer, agreed endpoint, or fixed denominator. People disagree about which gains matter and who should bear their costs. Improvements can produce new risks. Success under one metric can conceal regress under another.

That uncertainty is not a reason to abandon progress. It is a reason to replace faith with work. Choose goals openly. Measure several dimensions. Preserve the year and denominator. Look for people omitted by the average. Maintain gains that can be lost. Revise the route when evidence changes. Hope is most useful when it becomes a reason to act, not a reason to wait.

Sources for this chapter: Read [L014 - The Atlantic on the historical idea of progress](#) or its [L015 - archived duplicate](#). Study [L016 - Crawford's 1909 painting](#) as a visual argument, not as a literal data set. For the thermodynamics comparison, retain both scientific boundaries: the common entropy statement concerns an isolated system rather than merely a closed one, and physical entropy is not a direct measure of moral or political decline.

Final synthesis - Every indicator contains an argument

The progress bar, the airport walk, the dinosaur game, GDP, and a satirical painting all answer uncertainty by framing it. Some frames reveal useful information. Some improve an unavoidable wait. Some persuade. Some conceal. The scholar's task is to look through the frame without forgetting that it was built.

For any indicator, ask five things. What condition does it claim to measure? How is the value produced? What action does the display encourage? Who benefits if users believe it? What important experience remains outside it? These questions prevent a comfortable feeling of movement from replacing movement itself.

"Progress, not regress" is therefore not a prediction. It is a demanding programme. Real improvement must be defined, measured, distributed, maintained, and reconsidered. A bar that reaches one hundred percent can announce completion. Only people can decide whether the completed task made the world better.